Lesson 6 — Transducers: Sensors and Actuators

ATmega328P / Arduino Uno | Source: lecture 06 slides, AH1815, ADXL335, BC547, 2N7000, 1N4001 datasheets, "Applications of Diodes" | Builds on Lessons 3–5

WHAT YOU WILL BE ABLE TO DO AFTER THIS LESSON

- Define a transducer, and tell sensors from actuators and digital sensors from analog ones.
- Explain how each sensor in the lecture works and how to connect it to the microcontroller.
- Explain why actuators cannot be driven straight from a pin, and drive them with a **BJT**, a **MOSFET** or a **relay**.
- Calculate a base resistor, and explain the flyback diode.

GIVEN TO YOU IN THE EXAM

Nothing on the three exam sheets is specific to this lesson apart from the pin functions (digital pins, A0–A5, PWM pins 3, 5, 6, 9, 10, 11 on the Arduino sheet). **Everything below is "learn it".** Datasheets of the actual parts are in `Lecture Materials/06 - ...`; you only need the few numbers quoted here.

1. What is a transducer?

A **transducer converts energy (or a physical quantity) from one form to another**. Transducers sit at the **boundary** of an embedded system: they are how it touches the real world. They are grouped by the **direction** of conversion:

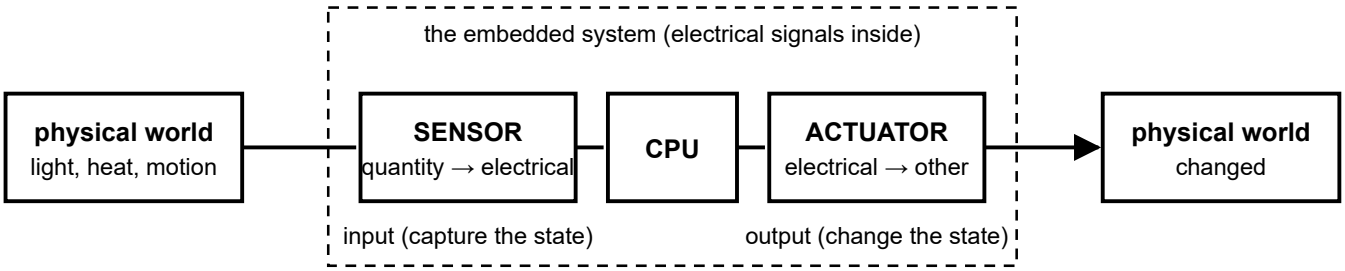


Figure 1. Sensors capture the state of the physical world as inputs; actuators change it.

SENSOR

Converts a physical quantity into an **electrical signal** (usually a voltage). Examples: LDR, LM35, reed switch, PIR, accelerometer.

ACTUATOR

Converts **electrical energy** into another form: light (LED, bulb), motion (motor, solenoid), heat/cold (Peltier), sound (buzzer).

2. Digital vs analog sensors

	Digital sensor	Analog sensor
Output	Two discrete voltages only: about 0 V or about 5 V ("yes/no": magnet present, motion seen)	A continuous voltage that follows the quantity (how bright, how hot, how tilted)
Connect to	Any digital I/O pin, read with `PINx` / `digitalRead`, or an interrupt pin	An ADC pin A0–A5, read with the ADC / `analogRead` (Lesson 5)
Lecture examples	Reed switch, Hall effect, PIR, flame	Water level, accelerometer, thumb joystick
Both	Sound, soil moisture, gas/smoke modules have an **Aout** (analog) pin **and** a **Dout** (digital) pin.	

How a module gives both Aout and Dout

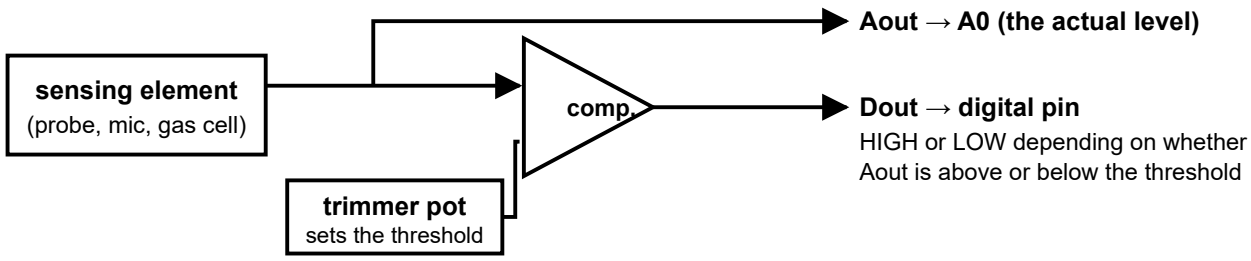


Figure 2. Inside an "analog/digital" module: a comparator (often an LM393 chip) compares the sensor voltage with a threshold you set with the small blue screw pot.

Use **Dout** when you only need "wet/dry", "loud/quiet", "gas/no gas" and want to set the threshold with a screwdriver. Use **Aout** when the program needs the level or must choose the threshold itself. Module pins are usually labelled **VCC (+5 V), GND, Aout, Dout**.

3. The digital sensors

Reed switch

Two thin metal contacts sealed in a glass tube with inert gas. **Normally open**; a **magnet** nearby magnetises the contacts so they attract and **close**. It is simply a switch operated by a magnet, so wire it exactly like a pushbutton: one side to the pin with a **pull-up**, the other side to GND.

Use: door/window alarms (magnet on the door, reed on the frame), position sensing, counting wheel turns.

Hall effect sensor (AH1815)

A semiconductor chip that produces a small voltage when a **magnetic field** passes through it (the Hall effect); built-in electronics turn this into a clean on/off output. The AH1815 is **omnipolar** (reacts to a north or south pole), runs on 2.5–5.5 V, and has an **open-drain output**: with a magnet the output is **pulled LOW**, otherwise it is "off" (floating). So it **needs a pull-up resistor** (internal pull-up or 10 kΩ to 5 V). No moving parts, so it is faster and lasts longer than a reed switch.

Pins (flat face towards you): VDD, GND, OUTPUT — check the datasheet drawing for your package.

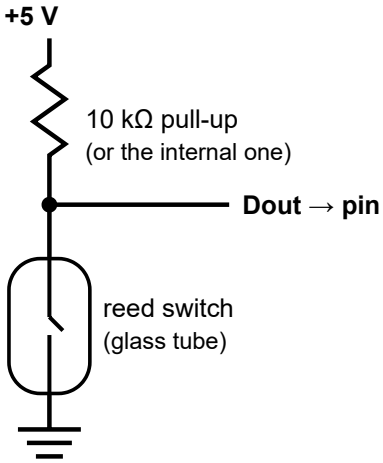


Figure 3. Reed switch with pull-up (same circuit for an open-drain Hall sensor: its output replaces the switch).

OPEN-DRAIN / OPEN-COLLECTOR, IN PLAIN WORDS

The output can only **connect the line to GND** (LOW) or **let go** (nothing). It cannot push the line HIGH by itself, so a pull-up resistor provides the HIGH. It lets the sensor work with any logic voltage.

PIR sensor (motion)

Pyroelectric **I**nfra**R**ed. Warm bodies give off infrared. A pyroelectric element produces a voltage when the IR falling on it **changes**; a white **Fresnel lens** splits the view into zones, so a person moving across zones causes changes. It detects **movement**, not presence: a person standing still is not seen. The module output goes **HIGH** (about 3.3 V, still read as HIGH) for a few seconds after motion. Modules have two trimmers (sensitivity and on-time) and need about **30–60 s to settle** after power-up. Pins: **+5 V, Dout, GND**.

Flame sensor

An IR photodiode/phototransistor tuned to the infrared emitted by a **fire**, with a comparator giving a digital output (often **LOW = flame**, check your module's LED). Use: fire alarms (the slide's example of a microcontroller system with binary inputs and outputs).

4. The analog sensors

Sensor	How it works	Connect / read
Water level	Exposed parallel copper traces. Water between them conducts; the more traces covered, the lower the resistance, so the output voltage rises with the water level.	Aout → A0. Power it only while reading (electrolysis corrodes the traces).
Accelerometer (ADXL335)	A tiny suspended mass inside the chip moves under acceleration; capacitive plates sense how far. Gives **three voltages**, one per axis X, Y, Z. At rest it measures **gravity**, so it senses **tilt**. Output is **ratiometric**: at 0 g each axis sits at half the supply (1.5 V at 3 V), about **300 mV per g**.	X, Y, Z → A0, A1, A2. The chip runs on 1.8–3.6 V; use a module with a regulator for 5 V (check yours).
Thumb joystick	Two potentiometers (X and Y) moved by the stick, plus a push switch when pressed down. Centre ≈ 2.5 V (ADC ≈ 512).	VRx → A0, VRy → A1, SW → digital pin with pull-up.
Sound	Microphone + amplifier: voltage follows the sound wave; Dout via comparator for "loud".	Aout → A0 or Dout → pin.
Soil moisture	Two probes in the soil: wet soil conducts better. Many modules give a **lower** Aout when wet; check by testing.	Aout → A0, or Dout.
Gas / smoke (MQ series)	A heated sensing layer whose resistance falls in the presence of gas. The heater needs a warm-up time and draws noticeable current.	Aout → A0, or Dout.
LDR, LM35, pot	Covered in Lesson 5 (voltage divider, 10 mV/°C).	A0–A5.

Accelerometer example: board flat: Z feels +1 g, X and Y 0 g. Tilt so X points down: X reads +1 g. With 3 V supply: X voltage = 1.5 V + 0.3 V × g. ADC on 5 V reference: 0 g ≈ 1.5 × 1024 / 5 = 307, 1 g ≈ 1.8 V ≈ 369.

5. Actuators, and why a pin cannot drive them directly

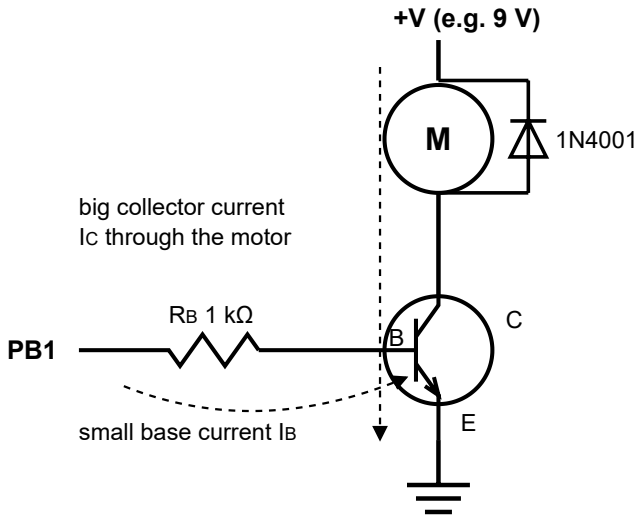
An I/O pin can supply only **5 V** and about **20 mA** (40 mA absolute maximum, and the whole chip about 200 mA). An LED (15 mA) is fine. But:

Actuator	Converts to	Needs
RGB LED	red, green, blue light; mixing gives other colours	Fine from pins: three LEDs in one package, one resistor and one pin each (PWM pins to mix shades, Lesson 8). **Common cathode**: shared leg to GND, pin HIGH = on. **Common anode**: shared leg to 5 V, pin LOW = on.
Incandescent bulb	light and heat	Higher voltage and current: transistor or relay.
DC motor	circular motion	Even small ones draw 100–500 mA, and more when starting: transistor/MOSFET + flyback diode, often its own supply.
Solenoid	straight-line motion (a plunger)	Higher voltage and current: MOSFET + flyback diode.
Thermoelectric (Peltier)	heat on one face, cold on the other	Amps of current: MOSFET or relay, separate supply.

THE RULE

The pin only **controls** a switch; a separate, stronger supply **powers** the load. The switch is a **BJT** transistor, a **MOSFET**, or a **relay**. The grounds of the two supplies must be joined, or the control signal has no reference.

6. Driving a load with a BJT (BC547)



GND of the 9 V supply joined to the Uno GND

Figure 4. Low-side switch: the load sits between +V and the collector; the emitter goes to GND.

How a BJT works: it has three legs, Base, Collector, Emitter. A **small current into the base** allows a **much bigger current from collector to emitter**. The ratio is the current gain $\mathbf{h_{FE}}$ (110–800 for BC547): $I_C = h_{FE} \times I_B$. As a switch we give **more** base current than needed, so the transistor is fully on (**saturated**: only about 0.2 V across C–E) or fully off (no base current).

Water picture: the base is a small tap controlling a big valve.

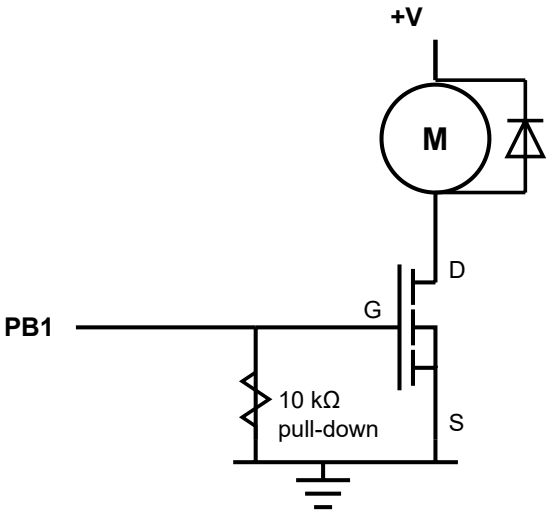
BASE RESISTOR CALCULATION

1. Load current I_C, e.g. relay coil 70 mA. 2. Minimum base current = $I_C / h_{FE(min)}$ = 70 / 110 = 0.64 mA. 3. Multiply by about 5 for hard saturation: $I_B \approx 3.2$ mA. 4. $R_B = (V_{pin} - V_{BE}) / I_B = (5 - 0.7) / 0.0032 = 1.3\ k\Omega \rightarrow$ use **1 kΩ** (gives 4.3 mA, still safe for the pin).

BC547 limits: I_C max **100 mA**, V_{CE} max 45 V. A bigger motor needs a bigger transistor or a MOSFET. **Pins (flat face towards you):** C, B, E.

Why "low side"? With an NPN the emitter must be at GND so the 5 V pin can push current into the base. The load goes on the "high" (+V) side.

7. Driving a load with a MOSFET (2N7000)



gate takes almost no current, so no gate resistor is needed for

Figure 5. N-MOSFET as a low-side switch.

How a MOSFET works: three legs, Gate, Drain, Source. It is controlled by **voltage**, not current: when the gate is more than the **threshold voltage** (0.8–3 V for 2N7000) above the source, a channel opens and current flows from drain to source. The gate is insulated, so it draws **almost no current** (only a tiny charge when switching).

The 10 kΩ pull-down keeps the gate at 0 V (MOSFET off) while the microcontroller is resetting and its pins are still floating inputs. Otherwise the motor might twitch at power-up.

2N7000 limits: about 200 mA continuous, 60 V. For motors of several amps use a "logic-level" power MOSFET (one fully on at V_{GS} = 5 V). **P-MOSFET**: the opposite type, used on the high (+V) side; gate pulled LOW to turn on.

	BJT (BC547)	MOSFET (2N7000)
Controlled by	base **current**	gate **voltage**
Needs	base resistor	pull-down (recommended)
Drop when on	~0.2 V ($V_{CE(sat)}$)	I × $R_{DS(on)}$ (a few Ω)
Best for	small loads, cheap	bigger loads, PWM, efficiency

8. The flyback diode: why every motor and relay needs one

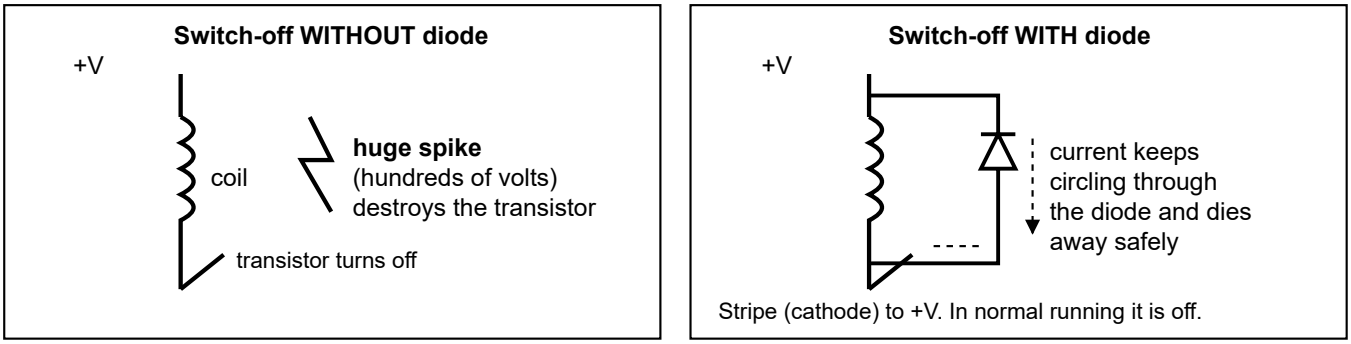


Figure 6. A coil (motor, relay, solenoid) stores energy in its magnetic field and resists any sudden change of current.

Why it happens: current through a coil cannot stop instantly. When the transistor suddenly opens, the coil forces the current to keep going by producing a very high voltage across the transistor. The **flyback (freewheeling) diode** placed across the coil, **cathode to +V**, gives that current a loop to circulate in until it dies out. It is "reverse-biased" (off) while the load runs normally. The 1N4001 (1 A, 50 V, ~0.7–1.1 V drop) is the standard choice. **Other diode uses** (web resource): rectifying AC to DC, protecting a circuit from a reversed battery, and LEDs themselves.

9. Switching with a relay

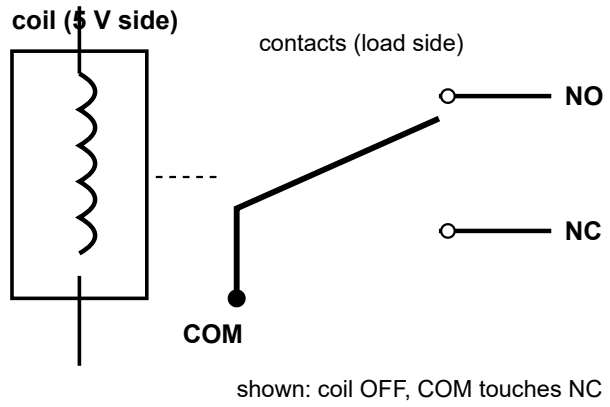


Figure 7. A relay: an electromagnet that moves a switch. COM = common, NO = normally open, NC = normally closed.

How it works: current through the **coil** makes a magnetic field that pulls the moving contact from **NC** over to **NO**. Coil off: a spring pulls it back. You hear a click.

Why use one: the coil circuit and the contact circuit are **electrically isolated**, so 5 V logic can switch **230 V AC** or large DC loads. The contact rating (e.g. 10 A 250 VAC) is printed on the case.

Driving it: the coil needs more current than a pin gives, so use the BJT circuit of section 6 with the coil as the load and a flyback diode across it. A **relay module** already contains the transistor, flyback diode, an indicator LED and screw terminals: you connect VCC, GND and IN (to the pin).

Wiring a lamp: live (through a fuse) → COM, NO → lamp, lamp → neutral. Lamp lights only while the coil is on. Use NC instead for "on unless activated". **Never build mains circuits on a breadboard.**

Limits: slow (milliseconds), so not for PWM; contacts wear out.

10. Worked design questions (exam style)

A. Door alarm with a reed switch

Spec: when the door opens, an LED (or buzzer) turns on and stays on until reset. **Design:** magnet on the door, reed switch on the frame, wired from **PD2 (pin 2)** to GND with the internal pull-up. Door closed: magnet near, reed closed, pin LOW. Door open: reed opens, pin goes HIGH. LED on **PB0 (pin 8)** through 220 Ω.

```
void setup()
{
  pinMode(2, INPUT_PULLUP);       // reed switch
  pinMode(8, OUTPUT);             // alarm LED / buzzer
}
void loop()
{
  if (digitalRead(2) == HIGH)     // door opened
    digitalWrite(8, HIGH);        // latch on
}                                 // reset button clears it
```

```
#include <avr/io.h>
int main(void)
{
  DDRB  |=  (1<<PB0);
  DDRD  &= ~(1<<PD2);
  PORTD |=  (1<<PD2);             // pull-up
  while (1)
    if (PIND & (1<<PD2))          // HIGH = open
      PORTB |= (1<<PB0);          // alarm on
}
```

B. Fan that switches on when it is hot

Spec: LM35 on A0; a 6–9 V DC motor turns on at 30 °C and off at 28 °C. **Design:** motor powered from the 9 V battery through a 2N7000 (or BC547 if the motor draws under 100 mA) on **PB1 (pin 9)**, flyback diode across the motor, 10 kΩ gate pull-down, grounds joined. Two thresholds (hysteresis, Lesson 5) stop it chattering at 29 °C. Pin 9 is also a PWM pin, so Lesson 8 can later control the **speed** (Exercise 6).

```
void setup() { pinMode(9, OUTPUT); }
void loop()
{
  float t = analogRead(A0) * 5.0 / 1024 * 100;   // LM35: degrees C
  if (t >= 30)      digitalWrite(9, HIGH);       // hot: fan on
  else if (t <= 28) digitalWrite(9, LOW);        // cool: fan off
  delay(500);
}
```

11. Build example: a motor driven by a BC547

This is component test 09 in the `Component Tests` folder (wiring and pictures there). Use the Lesson 3 section P method. What is new for actuators:

1. **Two supplies:** the motor runs from the 9 V battery, never from the Uno 5 V pin. Join the battery **−** to the Uno GND rail. Never connect the battery **+** to the Uno 5 V or the breadboard + rail used by the Uno.
2. **Order of wiring:** transistor first (check C, B, E), then 1 kΩ from the pin to the base, emitter to GND, motor between battery + and collector, diode across the motor (stripe to battery +). Connect the battery **last**, after checking.
3. **Test in steps:** blink sketch on the pin with an LED in place of the motor first (proves the transistor), then the motor.
4. **Signs of a fault:** Uno resets when the motor starts (motor on the Uno supply); motor always on (C and E swapped or transistor damaged); transistor hot (diode reversed, or motor current above 100 mA).

12. Common mistakes

Mistake	Result / fix
Motor or relay straight on a pin	Pin overloaded, chip may be damaged. Always a transistor/MOSFET.
No flyback diode on a coil	Spike kills the transistor, resets the chip. Diode across the coil, stripe to +V.
Grounds not joined	The transistor never switches. Join the load supply GND to the Uno GND.
No base resistor	Huge base current: pin and transistor damaged.
No pull-up on reed / open-drain Hall	Input floats, random readings.
Expecting PIR to see a still person	It detects movement only; also wait for warm-up.
3.3 V-only sensor on 5 V	Can be damaged (bare ADXL335). Check the module voltage.
Load current above the part's limit	BC547 > 100 mA or 2N7000 > 200 mA overheats. Pick a bigger part.

Cheat sheet

Sensors	
Digital	2 levels, digital pin. Reed, Hall, PIR, flame
Analog	continuous, A0–A5. Water level, accel., joystick
Both	Aout + Dout (comparator + trimmer). Sound, soil, gas
Reed / Hall	magnet → pin LOW (with pull-up)
PIR	movement → HIGH a few s; warm-up 30–60 s
ADXL335	0 g = Vs/2, ~300 mV/g, 3 axes

Actuator drivers	
Pin limit	5 V, ~20 mA (40 mA max)
BJT	$I_C = h_{FE}\, I_B$; $R_B = (5 - 0.7) / I_B$; BC547 ≤ 100 mA
MOSFET	gate voltage > threshold; 10 k pull-down; 2N7000 ≤ 200 mA
Flyback	diode across coil, stripe to +V
Relay	coil vs contacts isolated; COM / NO / NC
Always	separate load supply, grounds joined